



Cyber Security

Project 4

Industrial Training Kit

Batch: 2026 | Powered by DecodeLabs

WELCOME TO THE TEAM! 🚀

Step into the role of a Cybersecurity Analyst at DecodeLabs. Project 4 is your Optional Mastery Phase: The System Vulnerability Checklist. This track isn't about "simple checks"—it's about Proactive Hardening. Since this is an optional milestone, you can choose to claim your certificate now or complete this task to add "Risk Assessment" to your verified skills. By finishing this, you prove you can identify critical gaps through pure defensive logic.



Project 4: System Vulnerability Checklist



Goal:

Create a checklist to identify basic security vulnerabilities in a system.

Key Requirements:

- Check weak passwords
- Check software update status
- Identify unsafe user practices

Key Skills:

System security basics, risk
assessment

Project 4: Audited Systems

A Framework for Blue Team Defense

```
[SYS_INIT] Establishing baseline... [OK]  
[AUTH] Loading Blue Team protocols... [OK]
```


The Shift to Proactive Defense

The Legacy View

Security as a Product

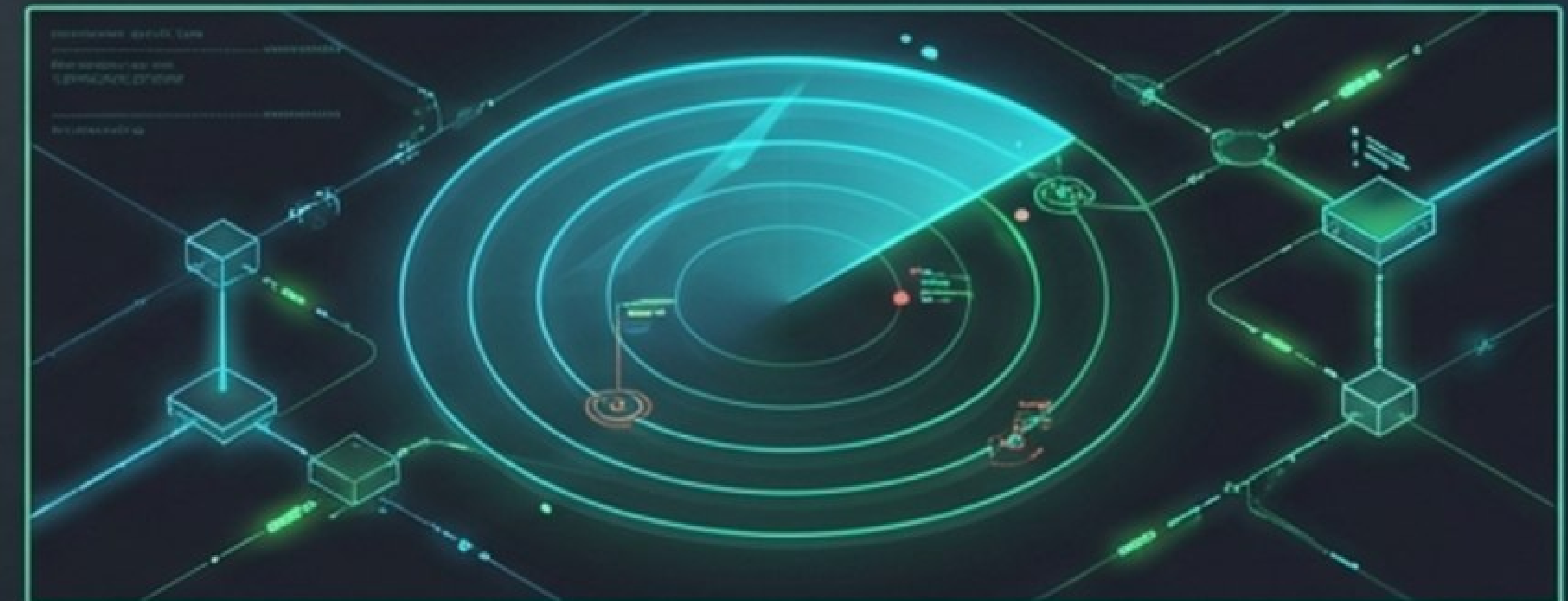


STATUS: STATIC DEFENSE [LEGACY_MODE]

- Defending single components
- Reactive remediation
- Assuming the perimeter is secure

The Blue Team Mindset

Security as a Continuous Process

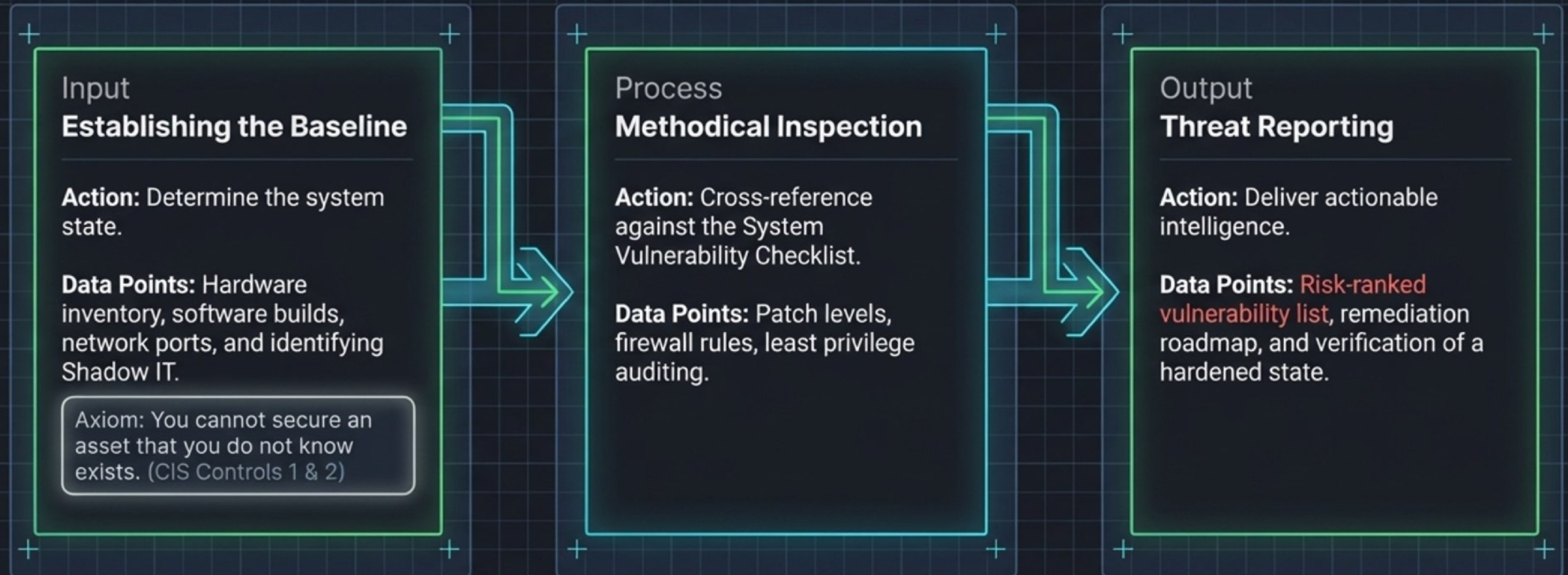


STATUS: ACTIVE SCAN [PROACTIVE_DEFENSE]

- Proactive hunting
- Assuming constant software decay
- Mitigating zero-day exploits
- Developing the Auditor's Eye

A system's integrity is dynamic. The security auditor is the essential gatekeeper for organizational resilience.

The Blueprint to Fortification



Checklist Step 1: The Identity Front Door

[SYSTEM_IDENTITY_CHECK: INITIATED]	Legacy Protocols	NIST 800-63B Standards	[MATRIX_TRANSLATION: ACTIVE]
Passwords	Complex requirements with forced periodic expiration. 	Strong passphrases. No forced expiration unless compromised. Focus on human memory and length. 	
Multi-Factor (MFA)	Out-of-band SMS or voice-delivered codes. (Vulnerable to interception and MFA fatigue). 	Phishing-Resistant Authenticators. Requires FIDO2/WebAuthn or Syncable Authenticators (Passkeys) tied to cryptographic device keys. 	
[SYSTEM_IDENTITY_CHECK: INITIATED] [SYSTEM_IDENTITY_CHECK: INITIATED]			[MATRIX_TRANSLATION: ACTIVE] [MATRIX_TRANSLATION: ACTIVE]

Checklist Step 2: Software Decay & Patch Management

Audit Objectives

- ✦ Verify automated OS updates are enforced.
- ✦ Identify unpatched browser builds and out-of-date antivirus databases.
- ✦ Hunt for Shadow IT (unapproved applications bypassing corporate restrictions).

The Auditor's Rule:

Delaying an update is explicitly accepting a risk.

System Risk

Time Since Last Update

Patch Applied



Checklist Step 3: Auditing the Human Perimeter

Step 3 of 4

[SYSTEM.AUDIT.PERIMETER: ENGAGED]

System Misconfigurations

[THREAT.VECTOR.ANALYSIS: ACTIVE]

The Guest Account Threat

Deprecated/High Risk

Disabling guest accounts prevents untrusted users from conducting basic reconnaissance or utilizing privilege escalation attacks.

Privilege Creep

Check for unchecked, lingering Admin group memberships. Enforce Least Privilege.



[SYSTEM.AUDIT.PERIMETER: ENGAGED]

Physical & Behavioral Hygiene

[THREAT.VECTOR.ANALYSIS: ACTIVE]

Physical Security

Verify screen locking timeouts, check for local credential storage (sticky notes, unencrypted files), and rogue USB peripherals.

AI-Driven Social Engineering

By 2025, deepfakes and cloned voices are corporate-scale threats. Auditors must verify employee reliance on secondary, trusted communication channels.

Checklist Step 4: Network and Endpoint Hygiene



The Shield (Inbound Traffic Controls)

Focus: The OS Firewall.

Action: Verify the firewall is active and blocking unauthorized inbound traffic.

Specific Callout: On macOS, ensure the Application Firewall (ALF) is active and Stealth Mode is enabled to prevent the system from responding to external reconnaissance pings.



The Vault (Data at Rest)

Focus: Local Disk Encryption.

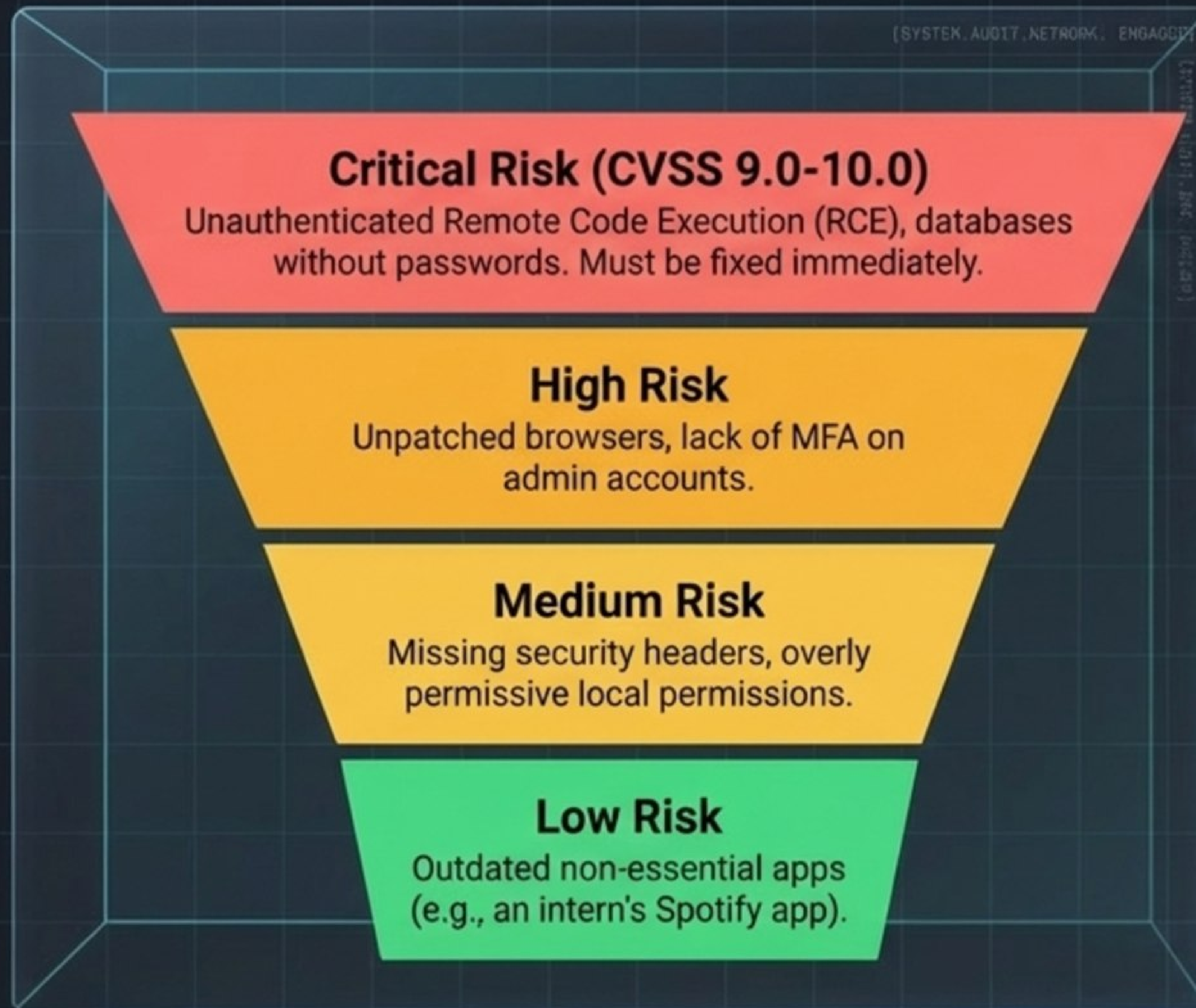
Action: Validate that the entire local disk is encrypted to protect data in the event of physical theft.

Tools: Verify FileVault is active on macOS and BitLocker with TPM is active on Windows.

The Auditor's Toolkit: Execution Matrix

Security Check	Windows (PowerShell)	macOS (Terminal - Zsh/Bash)
Verify Firewall	<code>Get-NetFirewallProfile</code>	<code>socketfilterfw --getglobalstate</code>
Check Encryption	<code>Get-BitLockerVolume</code>	<code>fdsetup status</code>
Audit Admin Rights	<code>Get-LocalGroupMember</code>	<code>dsccl . -read /Groups/admin</code>
Find Shadow IT	<code>Get-ItemProperty</code>	<code>ls /Applications</code>
Search for Updates	<code>Get-WUList</code>	<code>softwareupdate -l</code>

Categorizing Risk: The CVSS Funnel



CVSS 4.0 Metric Groups



Intrinsic Severity (Base)



Current Exploit Landscape (Threat)



Organizational Context (Environmental)

The Output: The Gatekeeper Rule

The Rule: To pass the foundational Cyber Security track at DecodeLabs, you must perform this exact 4-step checklist on your own primary computer.

The Philosophy: This is not a theoretical exercise. You cannot defend an enterprise if you cannot secure your own perimeter.

The Deliverable Requirement: Submission of a professional, 1-page Vulnerability Report detailing the specific findings from your personal machine audit.

Anatomy of the Vulnerability Report

Section 1: Flaws Found (The Diagnosis)

Requirement: Document at least three specific security flaws discovered using the CLI tools and CVSS scoring.

Section 2: Remediation Actions (The Treatment)

Requirement: Detail the exact steps taken to fix the flaws (e.g., enabling WPA3, deleting the Guest Account via terminal, updating a zero-day browser vulnerability).

Section 3: Hardened Verification (The Proof)

Requirement: Provide the final terminal outputs proving the system is now in a secure, hardened state (e.g., firewall active, disk encrypted).

```
[OK] FileVault is On.  
[OK] System Integrity Protection is enabled.
```


Enterprise Scale: The Big Leagues

FLOW: ASCENDING STANDARDS ALIGNMENT

STATUS: PASSING

DESTINATION: GLOBAL COMPLIANCE FRAMEWORKS

ISO 27001:2022

Maps directly to technological controls like Annex A 8.9 (Configuration Management) and 8.16 (Monitoring Activities).

SOC 2 (Common Criteria)

Maps directly to CC6 (Logical and Physical Access Controls) and CC7 (System Operations/Patching).

CIS Critical Security Controls v8

Maps to Control 1 & 2 (Asset Inventory).

FLOW: ASCENDING STANDARDS ALIGNMENT

DESTINATION: GLOBAL COMPLIANCE FRAMEWORKS

Base Level: Project 4
System Vulnerability Checklist
(Passwords, Patches, Hygiene)

Takeaway: Startups and large corporations use **legally binding versions** of this exact checklist to **prove their security posture** to global governments.

The Irreplaceable Auditor's Eye

The Core Message:

As we look toward the future, the audit process will be influenced by AI-powered scanners and automated platforms.

The Human Element:

Tools can identify that a port is open, but only a human auditor can determine if that open port is a business necessity or a dangerous exception.

Master these basics today. Build the posture of Continuous Assurance to manage the complex infrastructures of tomorrow. Stay one step ahead in the digital war.

CONCLUSION

The absolute best way to finish your time at DecodeLabs is by demonstrating a "Security First" mindset. As this project is entirely optional, you have already met the core requirements for your certification! We are incredibly proud of the progress you have made throughout this journey.

Please submit all your completed tasks to the portal for final verification. We wish you the absolute best in your future career—stay vigilant, keep learning, and keep securing the digital world!



THANK YOU

 +91 89330 06408

 decodelabs.tech@gmail.com

 www.decodelabs.tech

 GREATER LUCKNOW, INDIA

